

Carter Mitchell Betts

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EDUCATION

Ohio Northern University – <i>Ada, Ohio</i>	August 2023 – Present
Bachelor of Science in Mechanical Engineering – Robotics Concentration, Computer Science Minor - GPA: 3.96 Expected Graduation: May 2027	

PROFESSIONAL EXPERIENCE

Evans Research Fellow – <i>Ohio Northern University, Ada, Ohio</i>	May 2026 – August 2026
- Optimized COLMAP photogrammetry pipeline to achieve surface noise repeatable within 0.257 mm across 27 trials - Built and trained a PointNet++ deep learning segmentation model to automatically identify hand webspace in 3D scans with a mean intersection-over-union (mIoU) of 0.774 under 5-fold cross-validation - Developed Python model of collagen fiber networks to relate fiber crimp state to skin elastic modulus, validated against published experimental data	
Process Innovation Engineering Intern – <i>Howmet Aerospace, Cleveland, Ohio</i>	May 2025 – August 2025
- Cut projected lubrication costs by \$400,000+ on aluminum forging press through hands-on testing of lubrication dilution and delivery method - Engineered a custom viscometer from concept to completion, combining specialized hardware design, circuit development, and embedded programming to reduce calculation turnaround from roughly 7 days to less than 1 hour	
Product Integrity Engineering Intern – <i>Howmet Aerospace, Cleveland, Ohio</i>	May 2024 – August 2024
- Developed R scripts to automate critical lifecycle calculations and figure generation, reducing analysis and reporting turnaround by ~70% - Automated strain gauge drilling tests by designing, programming, and assembling a custom robotic testing rig, eliminating manual measurement error	

RESEARCH EXPERIENCE

Ohio Space Grant Consortium STEM Scholar	August 2026 – May 2027
- Designing low-footprint, smart microgravity exercise equipment for deep space missions on small crew vehicles - Selected to present research findings at 35th Annual OSGC Student Research Symposium	
Undergraduate Research Assistant – Multilayered Artificial Nail Biomechanics	March 2026 – May 2026
- Developed ANSYS finite element model to simulate loading on complex fingernail assembly - Performed convergence study to optimize element sizing and verify solution independence from mesh density - Analyzed layer-by-layer stress distributions to characterize mechanical behavior of nail bed composite structure	
Undergraduate Research Assistant – Syndactyly Surgical Planning	January 2024 – Present
- Performed finite element analysis on hand webspace through the use of ANSYS Parametric Design Language (APDL) - Built a Python-based embedded control and imaging system to drive a stepper motor, synchronize multi-camera capture, and reconstruct 3D geometry for surgical optimization	

PUBLICATIONS

- “Biomechanical Analysis of Multilayered Artificial Nail Systems: A Survey-Informed Finite Element Approach” – second author, ASME IMECE 2026, Paper No. IMECE2026-192805 (Accepted)
- “A Photogrammetry-Based Approach for Patient-Specific Modeling in Syndactyly Surgery” – fourth author, ASME IMECE 2025, Paper No. IMECE2025-166019; presented full technical session, Memphis, TN

LEADERSHIP

American Society of Mechanical Engineers – Vice President	April 2024 – April 2026
- Spearheaded a cross-organizational fundraiser that raised over \$1,700 for engineering initiatives - Led weekly meetings of 40+ engineering students to promote technical discussion and student engagement	
Sigma Pi International – Brotherhood Committee Head	August 2025 – August 2026
Coordinated chapter-wide events and managed the semester event calendar for 30+ members	

TECHNICAL SKILLS

Languages: Python, Java, MATLAB, C++, ANSYS Parametric Design Language (APDL), R

Simulation & CAD: ANSYS Mechanical, pyMAPDL, SolidWorks, OnShape

Machine Learning & 3D Vision: OpenCV, PyTorch, PointNet++, COLMAP, PyVista, PyMeshLab, 3DF Zephyr

Embedded Systems: Arduino, Raspberry Pi, Programmable Logic Controllers (PLC), LabVIEW, circuit integration

Fabrication: Soldering, 3D printing, MIG welding, machine and hand tools